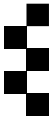
Foreign Keys and Referential Integrity Keys



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Candidate and Primary Keys

Candidate key

An attribute or set of attributes that uniquely determines the values of all the other attributes for a row in a relation.

Composite key

A key that contains two or more attributes.

Primary key

The candidate key selected to be the “official” key of a relationship.

- There is only one primary key per relation.
- The primary key may be a single key or a composite key.

Surrogate Keys

Surrogate key is a column of artificial data added to a relation to serve as the primary key.

Where does the value of a surrogate key come from?

A surrogate key value is normally assigned by the DBMS when a new row is created.

The assigned value never changes.

When would you use a surrogate key?

A surrogate key is used when the non-surrogate primary key would be too large and unwieldy.

More on Surrogate Keys

- The DBMS can then be used to assign a numeric value to PropertyID when a row is created (exactly how this is done depends upon which DBMS product is being used).
  - The AUTO_INCREMENT attribute can be used to generate a unique identity for new rows in MySQL.

- Surrogate key values are
  - artificial and have no meaning to the users.
  - normally hidden in forms and report

Example of Surrogate Keys

Consider the relation RENTAL_PROPERTY

RENTAL_PROPERTY (Street, City, State/Province, Zip/PostalCode, Country, Rental_Rate)

- For good performance a primary key should be short and, if possible, numeric.
- The primary key of RENTAL_PROPERTY is neither short not numeric.

RENTAL_PROPERTY (Property ID, Street, City, State/Province, Zip/PostalCode, Country, Rental_Rate)

Now, PropertyID is a **surrogate key** in this relation.

Foreign Keys

The attribute (or set of attributes) that is a primary key in one table that is then placed in a second table to form a relationship to the first table by storing linking values.

Example: Foreign Keys

DEPARTMENT.DepartmentName is the primary key of DEPARTMENT and EMPLOYEE.DepartmentName is a foreign key.

DEPARTMENT (DepartmentName, BudgetCode, ManagerName)

EMPLOYEE (EmployeeNumber, EmployeeLastName, EmployeeFirstName, *DepartmentName*)

Foreign keys express relationships between rows of tables.

Referential integrity constraint

A value constraint of a foreign key that states that no value can be placed in the foreign key unless it already exists as a primary key value in the linked table.

Example: Referential Integrity Constraints

SHIPMENT_3 (ShipmentNumber, ShipperName, DepartureDate, ArrivalDate, CountryOfOrigin, Destination, ShipmentCost, InsuranceValue, Insurer)

SHIPPER_FAX_NUMBER (ShipperName, ShipperFax)

WHERE SHIPPER_FAX.ShipperName must exist in SHIPMENT_3.ShipperName

SHIPPER_CONTACT (ShipperName, ShipperContact)

WHERE SHIPPER_CONTACT.ShipperName must exist in SHIPMENT_3.ShipperName